

Phenology-Guided Deep Learning with Uncertainty Quantification for Soybean Yield Prediction

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Abstract:

Accurate crop yield prediction is crucial for global food security, agricultural management, and economic planning in an era of increasing climate uncertainty. Deep learning approaches have shown promise in analyzing agricultural time series data for yield prediction but often lack explicit consideration of crop phenological development and uncertainty quantification. This chapter demonstrates an applied phenology-guided deep learning framework, integrating temporal sequences of satellite-derived phenological information with environmental variables to predict soybean yields. This approach builds upon established methodologies for incorporating phenology and probabilistic outputs in agricultural modeling. A primary objective of this chapter is to provide readers with a practical guide to implementing and comparing two distinct time series deep learning architectures - Long Short-Term Memory (LSTM) and one-dimensional Convolutional Neural Network (1D-CNN) - both extended with probabilistic outputs to quantify prediction uncertainties, complemented by accessible code examples. Using multi-temporal county-level data from Illinois, Indiana, and Iowa during 2008-2021, we evaluated the models' performance in capturing yield patterns across different

phenological stages. The LSTM model demonstrated superior performance in processing temporal sequences with an overall RMSE of 3.49 bu/ac and R^2 of 0.51, showing consistent accuracy across years. Feature importance analysis revealed that satellite-derived vegetation index (EVI2) and shallow soil moisture time series were the most influential predictors, while temporal analysis highlighted the critical importance of late reproductive stages for yield prediction. Spatial analysis of prediction uncertainties showed systematic patterns, with higher uncertainties in geographical boundary regions and areas with extreme yield predictions. By presenting this framework that explicitly accounts for temporal crop developmental stages and prediction reliability, this chapter offers practical insights and a methodological template for agricultural monitoring and decision-making systems. In line with NASA's Earth Science to Action strategy, we complement the methodological development with accessible code examples that demonstrate how long-term NASA satellite time-series observations can be transformed into operational decision-support tools for agricultural stakeholders.

Keywords: time series analysis, crop yield prediction, deep learning, phenology, uncertainty quantification

1. Introduction

The rising demand for food, driven by global population growth, changing dietary preferences, and increasingly frequent climate-related extremes, presents unprecedented challenges to modern agriculture. These challenges are further complicated by the depletion of natural resources and growing health and safety concerns in food production systems (Weiss et al., 2020). In this context, crop yield estimation has emerged as one of the most critical applications in modern agriculture, particularly given the increasing global focus on food security (Guan et al., 2017). Early and accurate crop yield estimation serves multiple crucial purposes: it helps reduce famine risks by providing timely assessments of food availability, enables policymakers to make informed import-export decisions for national food security, assists seed companies in evaluating hybrid performance across different environments, and helps farmers make better-informed management and financial decisions (Carletto et al., 2015; Sherrick et al., 2014).

Recent advancements in satellite technology have revolutionized agricultural monitoring capabilities through the provision of consistent, high-frequency temporal observations. Earth observation satellites, such as European Space Agency (ESA) Sentinel-2 and the National Aeronautics and Space Administration (NASA) MODIS (Moderate Resolution Imaging Spectroradiometer), provide systematic coverage with varying spatial and temporal resolutions. Sentinel-2 satellites deliver imagery at 10-60m resolution with 13 spectral bands and a five-day revisit period, while MODIS provides daily observations at 250-1000m resolution across 36 spectral bands (Drusch et al., 2012; Schaaf & Wang, 2015). The temporal dimension of satellite imagery is particularly crucial for crop yield estimation, as it captures the dynamic nature of crop development from planting to harvest. Vegetation indices derived from multi-temporal satellite

imagery, such as the Normalized Difference Vegetation Index (NDVI) and Enhanced Vegetation Index (EVI), provide valuable indicators of crop health and development status throughout the growing season (Rasmussen, 1992; Son et al., 2014; Weiss et al., 2020).

The emergence of deep learning techniques, combined with this increasing availability of satellite imagery time series, has opened new possibilities for crop yield estimation. Deep learning-based approaches combine satellite and ground-based time series data (e.g., vegetation indices, temperature, precipitation) with advanced neural network architectures to establish direct relationships between crop yields and vegetation greenness, as well as environmental factors captured in those time series (Chlingaryan et al., 2018; Kang et al., 2020, p. 202; Koirala et al., 2019). These approaches have demonstrated significant advantages in handling the temporal complexity of crop growth patterns and the high-dimensional nature of remote sensing data. Furthermore, deep learning frameworks enable the integration of multi-source information, including optical and radar imagery, along with meteorological variables, providing a comprehensive approach to yield prediction that accounts for various growth-influencing factors (Kang et al., 2020; Maimaitijiang et al., 2020; Muruganantham et al., 2022; van Klompenburg et al., 2020).

Multiple deep learning architectures have demonstrated effectiveness in crop yield estimation, each offering distinct capabilities in temporal data analysis. Long Short-Term Memory (LSTM) networks, a specialized form of Recurrent Neural Networks, have emerged as powerful tools for modeling temporal dependencies in yield prediction. These models excel at capturing long-term dependencies in sequential data, making them particularly suitable for processing time series of satellite and meteorological predictors throughout the growing season (Jiang et al., 2020; Tian et al., 2021; X. Wang et al., 2020; You et al., 2017). Convolutional

Neural Networks (CNNs), especially 1-dimensional CNNs (1D-CNNs) have also shown remarkable capability in extracting temporal features from satellite imagery sequences. Their inherent ability to learn hierarchical features from raw input data has proven valuable for processing multi-spectral satellite imagery (Barbosa et al., 2020; Kattenborn et al., 2021, 2021; Zhang & Diao, 2023). Additionally, hybrid architectures combining multiple neural network frameworks have emerged, such as CNN-LSTM models that integrate spatial feature extraction with temporal pattern recognition (Khaki et al., 2020; Sun et al., 2019). These hybrid approaches often achieve enhanced performance by leveraging complementary strengths of different architectures. More recently, architectures leveraging attention mechanisms, such as Transformer-based models originally successful in natural language processing, are also being explored in agriculture for tasks like crop yield prediction, demonstrating potential in capturing complex spatio-temporal dependencies from satellite and environmental data (Bi et al., 2023; F. Lin et al., 2023; Muruganantham et al., 2022; Xie et al., 2024).

Despite these advances, current deep learning approaches face two significant limitations. The first challenge lies in the lack of uncertainty quantification in yield predictions. Most existing models provide only deterministic predictions without considering the multiple sources of uncertainty inherent in agricultural systems (Lobell et al., 2009; Ma et al., 2021; X. Wang et al., 2020). These uncertainties stem from various sources: the inherent variability in weather patterns and their effects on crop growth, the potential noise and errors in satellite-based observations, the spatial heterogeneity of agricultural landscapes, and the model's structural limitations in capturing complex crop-environment interactions (Ma et al., 2021; Zhang & Diao, 2023). The absence of uncertainty estimates severely limits the practical utility of these models for agricultural decision-making, particularly in the context of risk assessment and management

under climate variability. While traditional statistical approaches often provide confidence intervals, incorporating uncertainty quantification into deep learning frameworks for crop yield estimation remains largely unexplored.

The second fundamental challenge lies in effectively incorporating the complex phenological patterns that occur throughout the growing season. Plant development follows distinct phenological stages, and the impact of environmental factors on final yield varies substantially depending on the growth stage at which they occur (Cohen et al., 2021; Guo et al., 2021; Jumrani & Bhatia, 2019). This temporal sensitivity is particularly evident in cases such as drought stress, where adverse conditions during critical growth periods (such as flowering or grain filling) can have disproportionately large effects on final yield compared to similar conditions during vegetative growth stages (Jumrani & Bhatia, 2018). Most existing deep learning approaches for yield estimation rely on simple temporal aggregation methods, typically using fixed calendar intervals such as weeks or months. This approach fails to account for the biological reality of crop development, as these arbitrary time periods often span multiple phenological stages or split important developmental phases. While some researchers have begun to address this limitation by incorporating phenological indicators into their models (Butts-Wilmsmeyer et al., 2019; Guo et al., 2021) or adjusting predictor variables based on key developmental stages (Bolton & Friedl, 2013), these solutions typically offer only partial representations of the complex developmental continuum of crop growth (Zhang & Diao, 2023). A comprehensive understanding of crop yield formation requires recognition that plants progress through numerous developmental stages, each with unique sensitivities to environmental conditions. The development of phenology-guided deep learning models that can effectively

integrate and analyze stage-specific predictors throughout the entire growing season represents a promising direction for advancing crop yield estimation accuracy.

In this chapter, we present phenology-guided probabilistic deep learning approaches that explicitly account for both crop developmental stages and prediction uncertainties in yield estimation. Our methods leverage the temporal resolution of satellite imagery and meteorological data while incorporating phenological information to better capture stage-specific crop responses to environmental conditions. The proposed framework not only provides yield predictions but also quantifies the associated uncertainties, offering valuable insights into prediction confidence levels across different growth stages and environmental conditions. Through the case study for soybean yield estimation, we demonstrate how these approaches can not only accurately predict yield with uncertainty estimates for risk assessment, but also better represent the biological reality of crop development, helping us understand the crop growth process. Crucially, this work advances the NASA Earth Science to Action framework by demonstrating how raw satellite observations can be fine-tuned and translated into actionable, real-world applications. Additionally, we provide practical implementation guidelines and code examples to help readers apply these methods to their own yield estimation challenges and bridge the gap between Earth observation data and on-the-ground decision-making.

2. Study Area and Data

The U.S. Corn Belt represents one of the world's most productive agricultural regions, contributing more than 81% of U.S. soybean and corn production in 2020. Within this region, Illinois, Indiana, and Iowa states serve as the study area for this case study. Together, these three states consistently rank among the top soybean-producing states in the U.S., contributing significantly to national soybean production. Their agricultural significance stems from their

ideal combination of environmental conditions: a humid continental climate characterized by warm summers and cold winters, relatively flat topography that facilitates large-scale mechanized farming, and deep, fertile soils developed from glacial deposits. These conditions, coupled with modern farming practices, make these states particularly suitable for intensive row crop agriculture, especially corn and soybean production.

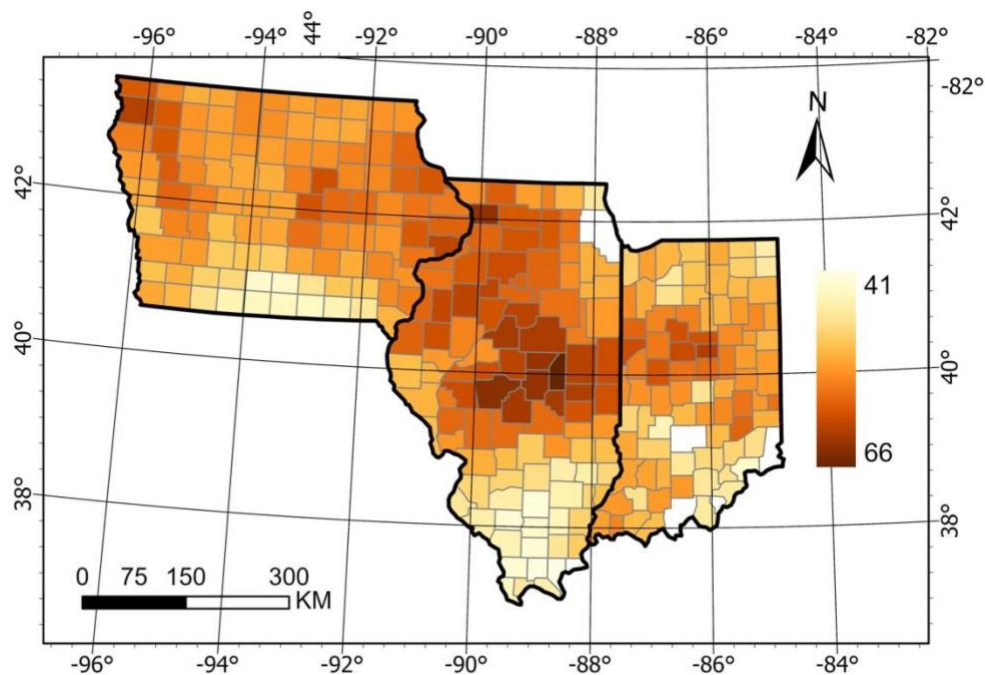


Figure.1 Study area showing average soybean yields (bu/ac) in Illinois, Iowa and Indiana from 2008-2021.

The study utilizes multiple datasets to analyze soybean yield patterns and their relationships with environmental variables. Soybean field locations are identified using the Cropland Data Layer (CDL), a 30-m resolution crop-specific land-cover dataset created annually for the US using satellite imagery and extensive ground measurements. County-level yield records spanning from 2008 to 2021 are obtained from the United States Department of Agriculture (USDA) National Agricultural Statistics Service (NASS) Database (USDA, 2022), encompassing 281 counties across the three states.

To effectively reflect the biological reality of crop development, the 14 satellite and meteorological predictors were deliberately selected to represent the three primary, biologically established drivers of soybean yield formation (Table 1): temporal vegetation vigor (canopy development and photosynthetic capacity), thermal time (heat accumulation and temperature stress), and water availability (soil moisture, precipitation, and atmospheric demand). To quantify the first of these drivers—vegetation vigor—the two-band Enhanced Vegetation Index (EVI2) serves as a direct indicator of soybean biomass. It is calculated as Eq.1:

$$EVI2 = 2.5 \times \frac{NIR - Red}{NIR + 2.4 \times Red + 1} \quad (1)$$

where Red and NIR represent the reflectance in red and near-infrared bands, respectively. EVI2 is extracted from the NASA Moderate Resolution Imaging Spectroradiometer (MODIS) MCD43A4 V6 Nadir Bidirectional Reflectance Distribution Function Adjusted Reflectance (NBAR) product (Sharma et al., 2018), which provides 500-meter nadir surface reflectance data on a daily basis. This daily 500-meter resolution product is particularly valuable as it minimizes atmospheric and soil background influences while maintaining enhanced sensitivity in high biomass regions (Schaaf & Wang, 2015). Recognizing that the impact of environmental variables on soybean yield fluctuates with the crop's developmental phases, this study also utilizes the MODIS MCD12Q2 Land Cover Dynamics product. This product provides annual, 500-meter pixel-level estimates of key vegetation phenology transition dates (Ganguly et al., 2010). These satellite-derived dates, which signify important shifts in soybean greenness and overall plant vigor, correspond well with established vegetative and reproductive milestones (Diao, 2019; Zhang & Diao, 2023). Segmenting the growing season based on these transition dates allows for a detailed analysis of how predictor variables change and influence yield within each distinct phenological stage of soybean development.

The selection of these MODIS products as the primary source for satellite-derived vegetation indices and phenology metrics was driven by several key considerations. Firstly, the MODIS archive offers a consistent and extensive data record spanning from the early 2000s, which fully covers our 2008-2021 study period — a longer temporal range than typically available from sensors like Visible Infrared Imaging Radiometer Suite (VIIRS) (Schaaf & Wang, 2015; Son et al., 2014). Secondly, the aforementioned MCD12Q2 product provides well-established, annual satellite-derived phenological transition dates essential for our phenology-guided framework, and these have been previously validated for soybeans in the study region (Diao, 2019; Zhang & Diao, 2023). Similarly, the MCD43A4 NBAR product is well-suited for robust EVI2 calculation, because its robust algorithm and 16-day compositing period used to generate the NBAR product inherently minimize cloud contamination and other atmospheric effects in the daily reflectance data (Z. Wang et al., 2018). While integrating data from multiple sensors could potentially enhance temporal data density, we opted for MODIS to leverage its long-term consistency, the direct availability of mature products tailored to our methodological needs, and to maintain consistency with foundational validation work (Jiang et al., 2020; You et al., 2017).

Environmental variables are sourced from various datasets. Daily maximum and minimum 2-meter air temperature and precipitation data are obtained from NASA Daymet V4 (Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 4; (Thornton et al., 2020)). Daily daytime land surface temperature (LST_Day), and daily nighttime land surface temperature (LST_Night) are extracted from the daily 1 km MODIS MYD11A1 product. Maximum Vapor Pressure Deficit (VPDmax) is derived from the daily 5km Parameter-elevation Regressions on Independent Slopes Model (PRISM) dataset. Additionally, both actual and

potential evapotranspiration and soil moisture at four depths are retrieved from the 3-hour 0.25 degree NASA Global Land Data Assimilation System (GLDAS).

Table.1. Description of the input data. The listed ‘Temporal Resolution’ refers to the native resolution of the source data; these predictors are subsequently temporally aggregated within six phenological stages for each county-year as detailed in Section 3.1.

Category	Variable	Spatial Resolution	Temporal Resolution	Source
Crop	Yield Record (bu/ac)	County-level	Annual	USDA Quick Statistic Database
	Cropland Map	30 m	Annual	NASS CDL Program
Phenology	Satellite Phenology	500 m	Annual	MODIS MCD12Q2
Vegetation Index	EVI2	500 m	Daily	MODIS MCD43A4
Heat-related	Maximum Air Temperature(°C)	1 km	Daily	Daymet V4
	Minimum Air Temperature(°C)			
Water-related	LST_Day (Kelvin)	1 km	Daily	MODIS MOD11A1
	LST_Night (Kelvin)			
	Precipitation(mm)	1 km	Daily	Daymet V4
	VPDmax (hPa)	5 km	Daily	PRISM
	Evapotranspiration (ET) (kg/m2/s)	0.25 °	3-Hour	GLDAS
	Potential ET (W/m2)			
	Soil Moisture 0_10cm (kg/m2)			
	Soil Moisture 10_40cm (kg/m2)			
	Soil Moisture 40_100cm (kg/m2)			
	Soil Moisture 100_200cm (kg/m2)			

3. Methods

This section presents Phenology-guided Neural Networks (Zhang & Diao, 2023) that integrate diverse crop growth and environmental predictors along soybean phenological development trajectories to estimate yields and associated uncertainties. The methodology

consists of two main components: phenology trajectory construction and deep learning-based yield estimation with uncertainty quantification.

To ensure spatial consistency, predictor datasets with native resolutions coarser than 500m were first resampled using nearest neighbor to align with the 500m grid of the MODIS data. The subsequent processing was conducted at this 500m pixel level for all soybean pixels, identified using the Cropland Data Layer (CDL) mask. For each year, the percentage of 30m CDL pixels classified as soybean within each 500m MODIS pixel's footprint was calculated. Only MODIS pixels where this soybean cover exceeded 90% were retained for feature extraction and analysis.

3.1. Phenology trajectory construction

The approach begins by dividing the crop growing season into six phenological stages using seven vegetation phenology transition dates obtained from the MODIS MCD12Q2 product. These transition dates include the onset of greenness (greenup), greenup midpoint (midgreenup), maturity, peak greenness (peak), senescence, greendown midpoint (midgreendown), and dormancy. Previous research has demonstrated strong correlations between these satellite-based phenological measures and National Crop Progress Report (CPR)-based soybean phenological stages in the US Corn Belt (Zhang & Diao, 2023). For example, satellite-based peak dates (mean: day of year 219) align with CPR-based pod-setting dates (mean: day of year 215), and dormancy dates (mean: day of year 288) correspond to harvest dates (mean: day of year 286) (shown in Table. 2). These strong correlations indicate that the satellite-derived transition dates effectively capture the key soybean growth stages and are suitable for dividing the growing season into phenologically meaningful periods.

Within each of the six identified phenological stages, environmental and satellite-based predictors are aggregated using stage-specific temporal boundaries for each pixel. Unlike traditional calendar-based methods that use fixed time intervals, this phenology-based aggregation approach adapts to the local timing of crop development, effectively capturing the temporal heterogeneity in crop growth patterns across different regions. The aggregation method varies by predictor type to best represent their influence on crop development. For precipitation, cumulative aggregation is employed within each phenological stage to capture the total water availability during each growth period. This cumulative approach aligns with the crop's response to water availability, as the total amount of precipitation during specific growth stages can significantly impact yield. In contrast, for other environmental variables such as temperature metrics, vapor pressure deficit, evapotranspiration, and vegetation indices, mean aggregation is used to characterize the average conditions within each stage. These variables influence or represent crop growth and development through their average levels rather than cumulative effects, making mean aggregation more appropriate. This stage-specific approach ensures that environmental conditions are properly aligned with actual crop developmental stages, regardless of spatial variations in timing. Following the stage-specific temporal aggregation of each predictor, the data were spatially aggregated to the county level. For each county polygon, we extract and average all pixels that fall within the county boundaries from each dataset. The final dataset comprises 84 features, representing 14 satellite and meteorological predictors across the six phenological stages.

Following these preprocessing steps, the data for each county and year is structured for input into the deep learning models. The target variable for prediction is the county-level soybean yield, as reported by the USDA National Agricultural Statistics Service (NASS) in

bushels per acre. The input features, derived from the 14 satellite and meteorological predictors (detailed in Table 1) aggregated across the six phenological stages, are organized as a temporal sequence. Specifically, for each county-year sample, this forms a sequence of six steps, where each step corresponds to one phenological stage and contains the vector of 14 predictor values for that stage. This results in an input array of shape (number of samples, 6 stages, 14 features per stage). Prior to model training, all input features derived from the training dataset (2008-2017) were standardized by removing the mean and scaling to unit variance; the same scaling parameters were then applied to the testing dataset (2018-2021).

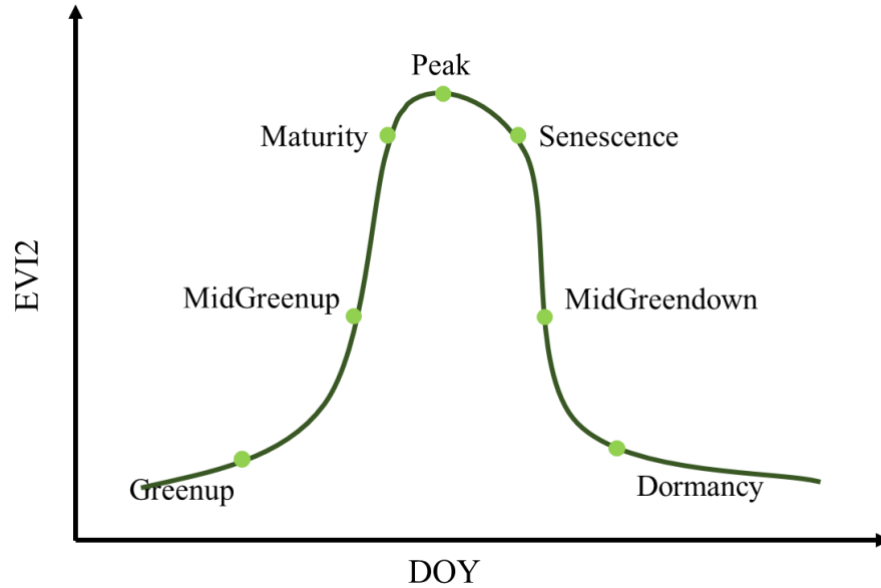


Fig.2. Division of growing season into six phenological stages based on satellite-derived transition dates.

Table.2. Comparison between satellite-derived and CPR-based soybean phenological stages in the U.S. Corn Belt (Zhang & Diao, 2023).

CPR-based transition date	Mean	Standard deviation	Satellite-based transition date	Mean	Standard deviation
Emerged	154	7	Mid-Greenup/Greenup	171/129	10/20
Blooming	198	6	Maturity	201	6
Setting pods	215	7	Peak	219	6
Dropping leaves	262	6	Mid-Greendown	262	5
Harvest	286	9	Dormancy	288	8

3.2. Time Series-based Deep Learning Models

For modeling the temporal dynamics of crop yield formation, we implement two state-of-the-art deep learning architectures specifically designed for time series analysis: Long Short-Term Memory (LSTM) networks and one-dimensional Convolutional Neural Networks (1D-CNN). These architectures are chosen for their proven capabilities in capturing temporal dependencies and sequential patterns in time series data (Kamilaris & Prenafeta-Boldú, 2018; Muruganantham et al., 2022; Zhu et al., 2017).

The LSTM architecture is particularly well-suited for analyzing the sequential nature of crop development throughout the growing season. By incorporating specialized memory cells and control mechanisms (forget gate, input gate, and output gate), LSTM networks can effectively model both long-term and short-term temporal dependencies in the phenological sequence of yield predictors. This capability is crucial for crop yield estimation, as crop development is inherently sequential, with events early in the season potentially influencing outcomes months later. The LSTM's ability to selectively remember or forget information through its gating mechanisms makes it especially effective at identifying which phenological stages and conditions are most critical for final yield outcomes (T. Lin et al., 2020; Schwalbert et al., 2020; Tian et al., 2021).

The 1D-CNN approach offers a complementary method for processing temporal sequences of crop and environmental data. Unlike traditional CNNs that process spatial data, our 1D-CNN implementation specifically targets the temporal dimension of the yield predictor sequences. The network architecture consists of multiple convolutional blocks, where each block contains convolutional layers applying one-dimensional filters along the temporal axis. These layers systematically extract temporal features at different scales, while subsequent pooling layers help

identify the most significant temporal patterns. This hierarchical processing is particularly effective for detecting stage-specific characteristics and their relationships to final yield. The 1D-CNN's ability to automatically learn relevant temporal features makes it especially valuable for identifying critical periods and conditions during the growing season that influence yield outcomes (Ben Hamida et al., 2018; You et al., 2017; Zhong et al., 2019).

Specific architectures for both LSTM and 1D-CNN models were developed through a structured manual hyperparameter optimization process. This involved iteratively experimenting with key parameters within predefined ranges: the number of LSTM or convolutional layers was selected from [1, 2, 4]; the number of units in LSTM layers from [50, 100, 200]; and the number of filters for convolutional layers from [16, 32, 64, 128]. The final architectures, detailed subsequently (e.g., LSTM with 100 units, 1D-CNN with specific filter counts), were chosen based on their performance on a held-out validation set and considerations of model complexity. These models were implemented using Keras 3.5 in Python 3.10. The LSTM network consists of an input layer that accepts the temporal sequence of phenological predictors, followed by an LSTM layer with 100 units to capture temporal dependencies. The extracted temporal features are then processed through a dense layer with 100 neurons and ReLU activation function for final yield prediction. The 1D-CNN architecture comprises three convolutional blocks for temporal feature extraction. The first two blocks each contain a Conv1D layer with 32 filters and kernel size of 3, followed by max pooling layers with pool size of 2. The third block consists of a Conv1D layer with 16 filters. All convolutional layers use ReLU activation and 'same' padding to preserve temporal sequence length. The extracted features are flattened and processed through a dense layer with 100 neurons and ReLU activation. Both architectures are designed to

efficiently process the temporal sequences of phenological-stage data while maintaining the ability to capture complex patterns in crop development throughout the growing season.

3.3. Uncertainty Quantification

To quantify yield estimation uncertainty, both models are extended to output Gaussian distributions characterized by mean ($\hat{y}$) and standard deviation (σ) parameters (shown in Fig.3). This probabilistic approach accounts for observation noise and variability in yield predictor inputs. The models are trained using a negative log-likelihood loss function (Kendall & Gal, 2017):

$$L(W) = \sum_{i=1}^N \left(\frac{1}{2\hat{\sigma}_i^2} \|\hat{y}_i - y_i\|^2 + \frac{1}{2} \log \hat{\sigma}_i^2 \right) \quad (2)$$

where $\hat{y}_i$ and y_i represent the estimated mean yield and observed yield for sample i , respectively, N is the number of training samples, and $\hat{\sigma}_i^2$ denotes the estimated variance. This probability modeling framework enables simultaneous estimation of yield values and their associated uncertainties, providing a more comprehensive understanding of model predictions. The standard deviation output (σ) serves as a measure of prediction uncertainty, offering insights into the model's confidence in its yield estimates across different conditions and locations. This approach primarily aims to capture aleatoric uncertainty, which arises from inherent noise or variability in the data and the observation process. While epistemic uncertainty, stemming from model ignorance due to limited data or model misspecification, can be estimated through methods like Bayesian neural networks or deep ensembles (Hüllermeier & Waegeman, 2021; Zhang & Diao, 2023).

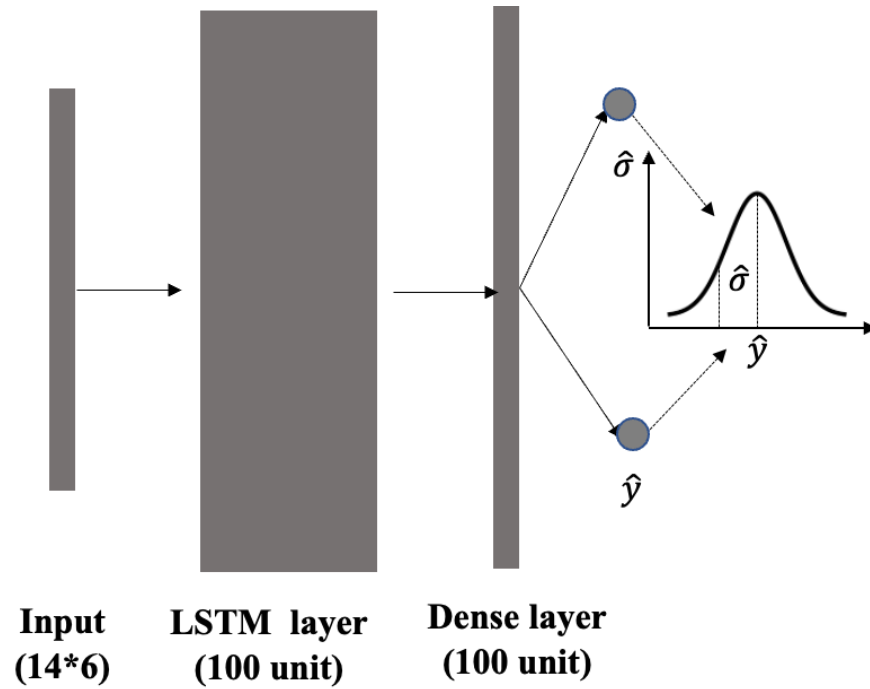


Fig.3. Architecture of probabilistic LSTM model for yield prediction with uncertainty quantification.

3.4. Feature importance evaluation

To understand the relative contributions of different environmental variables and phenological stages in yield estimation, we employ the permutation importance feature selection method (Breiman, 2001; Shahhosseini et al., 2020; X. Wang et al., 2020). This non-linear model inspection approach evaluates feature importance by measuring changes in model performance when specific features are randomly shuffled in the dataset. The permutation importance analysis is conducted using the following steps:

- Shuffle each predictor variable across all samples while keeping other variables unchanged.
- Evaluate the model's performance (e.g., coefficient of determination) with the shuffled dataset.
- Calculate the difference in performance between the original and shuffled datasets.

- Repeat steps 1-3 for each predictor variable or phenological stage.
- Rank the variables or stages based on their importance scores (i.e., the performance difference).

A larger decrease in model performance after shuffling indicates greater importance of that variable. We apply this analysis to each environmental and satellite-derived predictor (including temperature metrics, precipitation, vapor pressure deficit, evapotranspiration, and vegetation indices) at each phenological stage. This stage-specific analysis enables us to understand not only which variables are most influential for yield prediction but also when during the growing season their influence is most significant.

3.5. Model Training and Evaluation

The training process utilizes county-level data spanning from 2008 to 2017, with model performance evaluated on subsequent years (2018-2021) to assess temporal generalization capability. To optimize model performance, we employ the Adam optimizer with a learning rate of 0.0001. Training was conducted for a maximum of 1000 epochs. To prevent overfitting, an early stopping mechanism is implemented that monitors validation loss, halting the training process when no further improvement is observed. Model performance is evaluated using two widely adopted metrics in yield estimation: the Root Mean Square Error (RMSE) and coefficient of determination (R^2). These metrics are calculated as follows:

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (3)$$

$$R^2 = 1 - \frac{\sum_{i=1}^N (\hat{y}_i - y_i)^2}{\sum_{i=1}^N (\bar{y} - y_i)^2} \quad (4)$$

where N denotes the number of samples; y_i is the observed yield; $\hat{y}_i$ is the estimated yield; $\bar{y}$ is the average yield observation. RMSE provides a measure of the absolute prediction error in the same units as the yield measurements, while R^2 indicates the proportion of yield variance explained by the model, with values ranging from 0 to 1.

To provide a baseline for evaluating the deep learning models, a Gaussian Process Regression (GPR) model was implemented. GPR is a non-parametric, Bayesian approach to regression that can provide uncertainty estimates along with its predictions (Han et al., 2020; Martínez-Ferrer et al., 2021; Williams & Rasmussen, 2006; You et al., 2017). The GPR model utilized the same 84 phenology-guided features (14 predictors across 6 phenological stages) as the deep learning models. To accommodate the GPR's input requirements, the time-series structure of the data for each county-year sample (originally shape: 6 timesteps $\times$ 14 features) was flattened into a single feature vector of 84 elements. This ensures the GPR uses the same core information as the DL models, though without explicitly modeling the temporal sequence. The GPR model was implemented using Scikit-learn's GaussianProcessRegressor. A composite kernel was chosen, consisting of a constant kernel multiplied by a Radial Basis Function (RBF) kernel.

4. Results and discussion

4.1. Model Performance Comparison

Both LSTM and 1D-CNN models demonstrated strong capabilities in estimating soybean yields across the U.S. Corn Belt region during the testing period (2018-2021), with the LSTM model showing superior overall performance. As shown in Table 3, the LSTM model achieved an overall RMSE of 3.49 bu/ac and R^2 of 0.51, calculated on the pooled set of all predictions and

observations from the test years (2018-2021), while the 1D-CNN model recorded an RMSE of 3.73 bu/ac and R^2 of 0.44. These results indicate that both deep learning architectures can effectively capture the complex relationships between temporal environmental patterns and crop yield formation. The GPR baseline model, using the flattened phenology-guided features, achieved an overall RMSE of 3.75 bu/ac and an R^2 of 0.44 (Table 3). Compared to the deep learning models, the GPR performed comparably in terms of overall RMSE and R^2 with 1D-CNN. However, the LSTM model's overall RMSE of 3.49 bu/ac was lower than the GPR, and its R^2 of 0.51 was higher. This suggests that the deep learning models, particularly LSTM, were better able to leverage the sequential nature of the phenology-guided features, which was lost to the GPR due to feature flattening.

The LSTM model exhibited particularly consistent performance across all testing years. As illustrated in Fig. 4, the model maintained stable prediction accuracy with RMSE values ranging from 3.30 to 3.65 bu/ac and R^2 values between 0.38 and 0.56. The model achieved its highest prediction accuracy in 2020 (RMSE = 3.30 bu/ac, R^2 = 0.56), demonstrating its ability to effectively capture yield patterns during optimal growing conditions. Even in 2019, a year marked by widespread flooding and significantly delayed planting across the U.S. Corn Belt, the LSTM maintained relatively robust performance (RMSE = 3.53 bu/ac, R^2 = 0.38), indicating its ability to handle varying environmental conditions (Nafziger, 2019; Rippey, 2019; Schnitkey & Zulauf, 2019; Westcott, 2010).

The 1D-CNN model also demonstrated strong predictive capabilities, particularly in years with favorable growing conditions. As shown in Fig. 5, the model achieved excellent performance in both 2018 (RMSE = 3.45 bu/ac, R^2 = 0.48) and 2020 (RMSE = 3.29 bu/ac, R^2 = 0.56), matching or slightly exceeding the LSTM's performance in these years. These results

highlight the 1D-CNN's potential for high-accuracy yield estimation when environmental conditions follow typical patterns.

Analysis of the scatter plots in Fig. 4 and Fig. 5 reveals that both models show similar prediction patterns across the yield range, with slightly better performance in the moderate yield range (45-60 bu/ac) where most observations are concentrated. Both architectures showed minor tendencies to underestimate extremely high yields (>65 bu/ac), a common challenge in crop yield prediction models noted in previous studies (Jiang et al., 2020; Zhang & Diao, 2023). The primary difference between the two models lies in their temporal stability. While the LSTM maintained consistent performance across years, the 1D-CNN showed more variable results, particularly in 2019 and 2021 where its R^2 values decreased to 0.07 and 0.12, respectively (Table 3). This difference can be attributed to the LSTM's architecture, which is specifically designed to capture long-term dependencies in sequential data through its memory mechanisms. These mechanisms allow the LSTM to better account for the cumulative effects of environmental conditions throughout the growing season, leading to more stable predictions across different years and conditions. Both models demonstrate the potential of deep learning approaches for crop yield estimation, with each showing particular strengths in different conditions. The LSTM's superior overall performance and temporal stability make it particularly suitable for operational forecasting systems where consistent reliability is crucial. However, the 1D-CNN's strong performance in optimal conditions suggests it could be valuable in specific applications, particularly in regions with stable growing conditions or as part of ensemble prediction systems. These findings contribute to the growing body of evidence supporting the effectiveness of deep learning approaches in agricultural applications, while also highlighting the importance of careful model selection based on specific use case requirements.

When implementing phenology-guided approaches, practitioners should be aware that this study does not include direct comparisons with calendar-based temporal aggregation methods (e.g., monthly or weekly averages). While our results demonstrate strong performance with phenology-guided features, readers interested in implementing these methods should consider testing both approaches in their specific contexts. The phenology-guided framework offers the theoretical advantage of aligning data aggregation with biological crop development, but the magnitude of this benefit compared to simpler calendar-based methods may vary depending on the specific crop, region, and available phenological data quality.

Table.3. Performance metrics of LSTM and 1D-CNN models for soybean yield prediction (2018-2021). The “Overall” metrics are calculated from predictions pooled across all test years (2018-2021).

	LSTM		CNN		GPR	
	RMSE (bu/ac)	R2	RMSE (bu/ac)	R2	RMSE (bu/ac)	R2
2018	3.49	0.47	3.45	0.48	-	-
2019	3.53	0.38	3.84	0.26	-	-
2020	3.3	0.56	3.29	0.56	-	-
2021	3.65	0.47	4.39	0.23	-	-
Overall	3.48	0.52	3.73	0.44	3.75	0.44

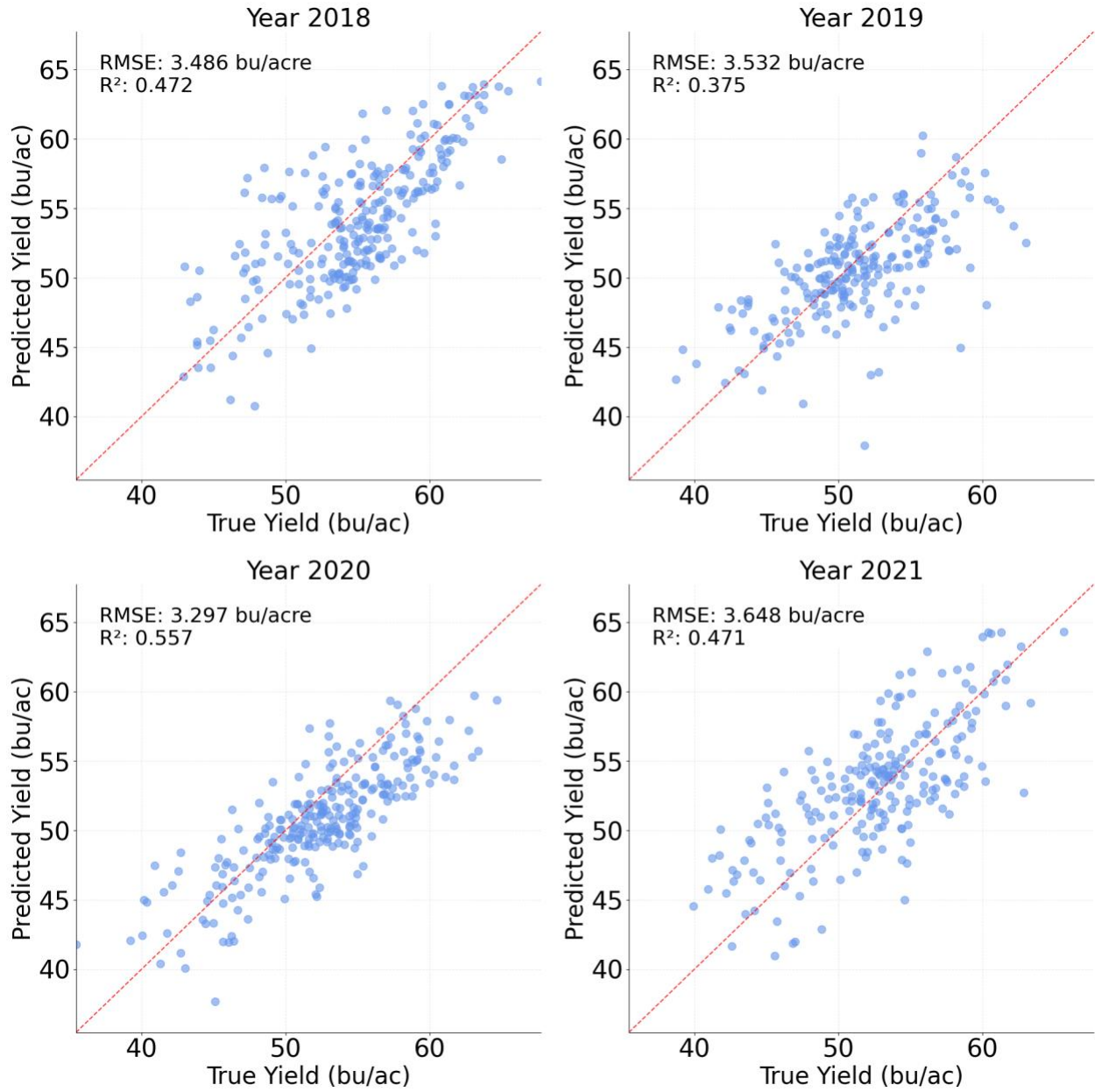


Fig.4. Scatter plots of observed versus predicted soybean yields using LSTM model for years 2018-2021. Each point represents a county-level yield prediction.

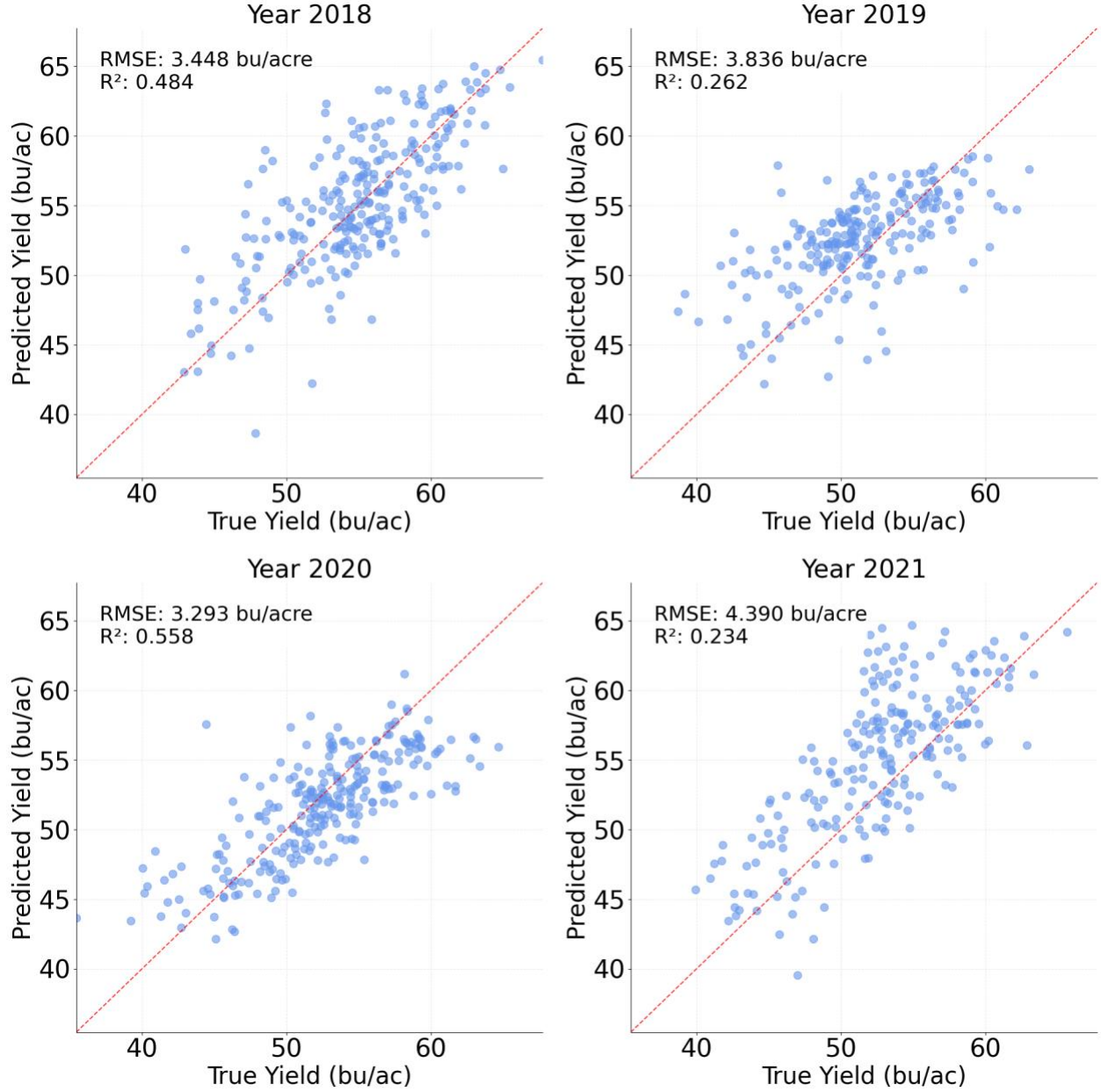


Fig.5. Scatter plots of observed versus predicted soybean yields using 1D-CNN model for years 2018-2021. Each point represents a county-level yield prediction.

4.2. Uncertainty Quantification Analysis

Both LSTM and 1D-CNN models were extended to provide probabilistic predictions with uncertainty estimates, enabling assessment of prediction reliability across different yield ranges and growing conditions. Figs. 6, 7 and 8 illustrate the predicted versus observed yields

with associated 95% confidence intervals for the LSTM, 1D-CNN and GPR models, respectively, across the testing period (2018-2021). The LSTM model demonstrates well-calibrated uncertainty estimates, with confidence intervals that adapt systematically to different yield ranges. In the moderate yield range (45-60 bu/ac), where most observations occur, the confidence intervals remain relatively narrow at approximately ± 3.5 bu/ac. These intervals widen appropriately for more extreme yield values, reflecting increased prediction uncertainty in less common conditions. Notably, the actual yields consistently fall within the predicted confidence bounds across all testing years, suggesting reliable uncertainty quantification. The 1D-CNN model exhibits similar patterns in uncertainty estimation but with generally wider confidence intervals. The average width of these intervals in the moderate yield range is approximately ± 4.2 bu/ac, increasing more substantially for extreme yields compared to the LSTM model. This pattern aligns with the model's more variable performance across years and indicates effective self-assessment of prediction reliability. In contrast, GPR shows larger confidence intervals across yield ranges.

To quantitatively assess the reliability of the probabilistic predictions from all models, we evaluated the 95% prediction interval coverage on the test set. The Gaussian Process Regression (GPR) baseline model achieved a 95% CI coverage of 85.43%. In contrast, the deep learning models exhibited significantly lower coverage: the LSTM model's 95% CIs covered only 45.09% of the true county-level soybean yields, and the 1D-CNN model's coverage was 43.34%. The notably low coverage rates for the LSTM and 1D-CNN indicate that their prediction intervals, derived from the learned standard deviations, are considerably too narrow. The underestimation of uncertainty by the DL models could stem from several factors. Their learned variance primarily targets aleatoric uncertainty and may not sufficiently account for

epistemic uncertainty, which the GPR model might capture more effectively through its Bayesian formulation. Other factors could include model misspecification for the DL models or the limitations of assuming a single Gaussian predictive distribution without further calibration. A more rigorous Uncertainty Quantification (UQ) analysis would involve generating standard diagnostics such as Prediction Interval Coverage Probability (PICP), calibration plots, and Probability Integral Transform (PIT) histograms. Furthermore, evaluating probabilistic metrics like the Continuous Ranked Probability Score (CRPS) and applying post-hoc calibration methods would be essential next steps to produce more reliable and trustworthy uncertainty estimates.

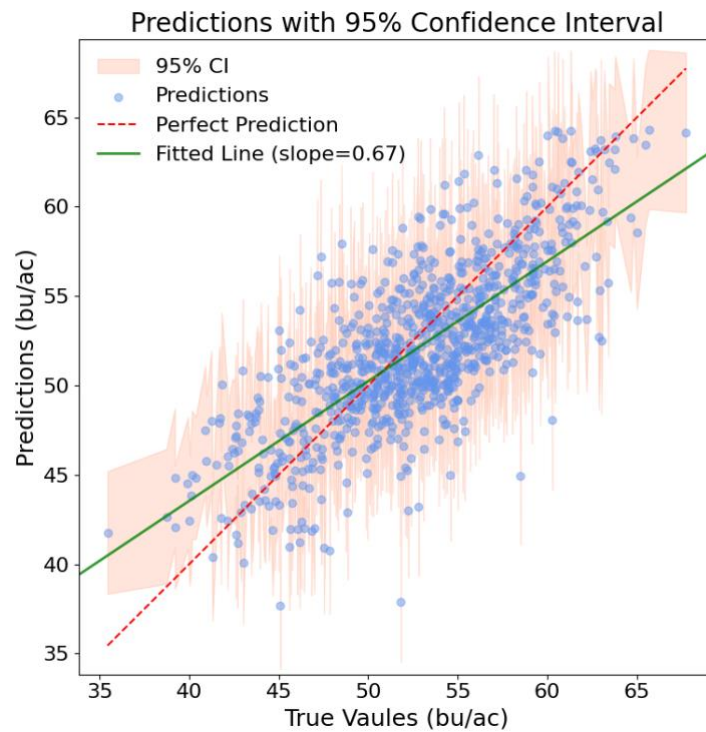


Fig.6. LSTM model predictions with 95% confidence intervals for soybean yields across Illinois, Iowa and Indiana (2018-2021). The shaded area represents prediction uncertainty.

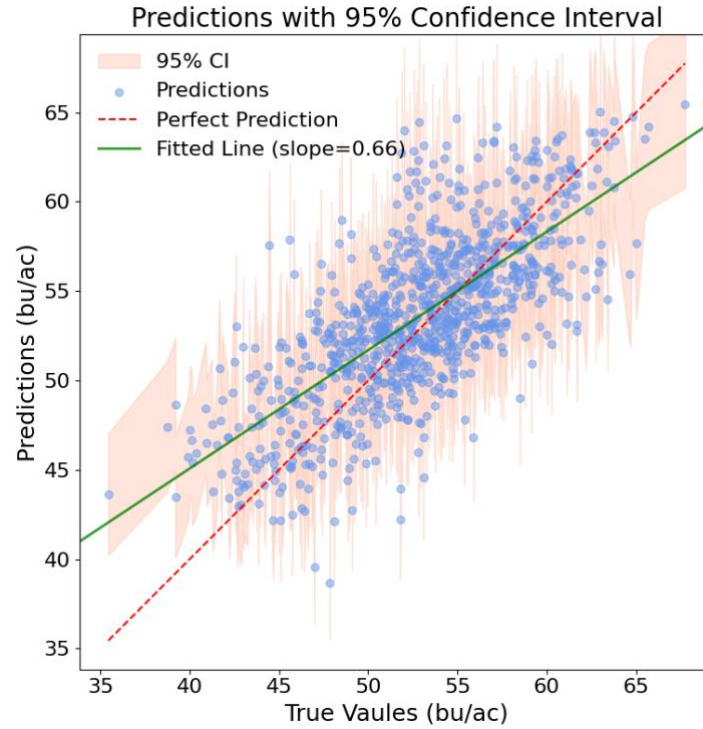


Fig.7. 1D-CNN model predictions with 95% confidence intervals for soybean yields across Illinois, Iowa and Indiana (2018-2021). The shaded area represents prediction uncertainty.

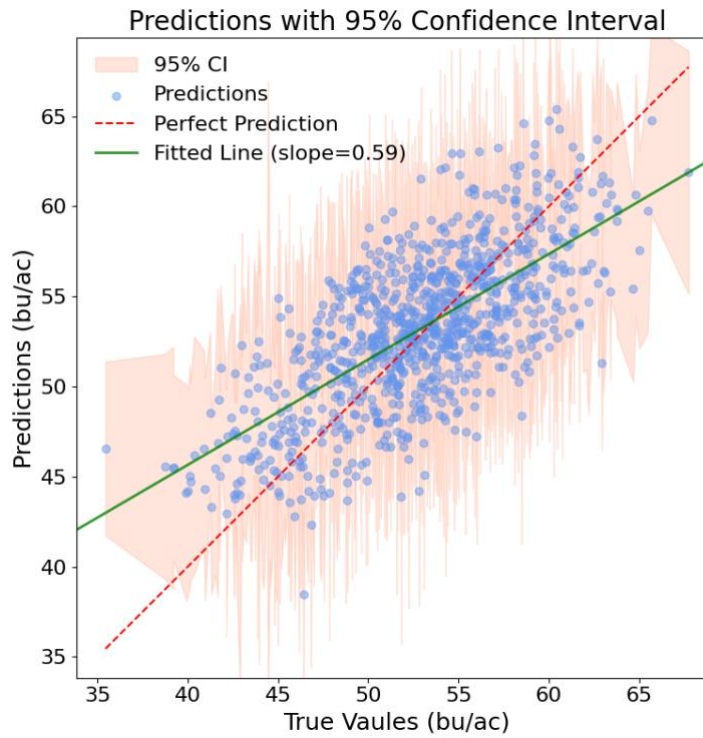


Fig.8. GPR model predictions with 95% confidence intervals for soybean yields across Illinois, Iowa and Indiana (2018-2021). The shaded area represents prediction uncertainty.

Given the LSTM model's superior performance, we conducted detailed spatial analysis of its prediction uncertainties. Fig. 9 illustrates the spatial distribution of LSTM yield predictions and their associated uncertainties across the study region from 2018 to 2021. The analysis reveals systematic geographical variations in prediction confidence, with notably higher uncertainties in counties along the study area boundaries. These elevated uncertainties typically occur in regions where environmental conditions differ markedly from those in the core agricultural areas of the U.S. Corn Belt, suggesting that the model's confidence decreases when encountering conditions that deviate from the primary training distribution patterns.

The spatial distribution of uncertainties also shows a strong correlation with predicted yield levels. Regions with predicted yields exceeding 60 bu/ac generally exhibit higher prediction uncertainties, particularly in the high-yielding areas of central Illinois and Iowa. This pattern aligns with the confidence interval analysis in Fig. 6, where prediction uncertainty increases for higher yield values, likely due to the relative scarcity of high-yield observations in the training data and the greater complexity in predicting exceptional crop performance.

While the overall spatial pattern of uncertainties remains consistent across the study period, the magnitude of uncertainties varies by year. For instance, 2019 shows generally higher uncertainty values across the region compared to other years, likely reflecting the challenging growing conditions experienced during that season. Despite these annual variations, the geographical pattern of higher uncertainties in boundary regions and high-yielding areas persists, suggesting that these patterns are inherent to the model's prediction capability rather than artifacts of specific growing conditions.

The observed patterns in prediction uncertainty reveal important characteristics of the LSTM model's performance across different spatial and temporal contexts. The systematic

variation of uncertainties with yield levels demonstrates the model's ability to identify conditions where its predictions may be less reliable. This is particularly evident in predictions of extreme yields and in regions along the geographical boundaries of the study area, where the model appropriately indicates lower confidence in its estimates. The spatial distributions of uncertainty also highlight the model's varying reliability across the study region. The consistent patterns of higher uncertainties in specific geographical areas and during particular growing conditions provide valuable insights into the model's limitations. These patterns suggest that while the model performs reliably across most of the U.S. Corn Belt, its predictions should be interpreted more cautiously in regions where environmental conditions deviate significantly from those in the core agricultural areas.

To further evaluate the LSTM model's performance, Figure 10 presents a time series comparison of the average actual soybean yield versus the average LSTM-predicted soybean yield for the 2018-2021 test period. The error bars in this figure represent ± 1 standard deviation of the county-level actual yields (or predictions) around their respective regional means, illustrating the spatial variability across counties each year. The plot of regional averages indicates that the LSTM model's predictions generally capture the directional trend of the observed year-to-year fluctuations in mean yield. For example, both actual and predicted mean yields show a decrease from 2018 to 2019, followed by a general rise by 2021. However, the model tended to underestimate the average regional yield in 2018, 2019, and more substantially in 2020, while slightly overestimating it in 2021. The error bars provide important insights into the spatial variability of yields. Actual county-level yields exhibited considerable spatial heterogeneity, and this degree of variation among counties fluctuated annually; for instance, 2019 showed relatively less spatial variation in actual yields compared to other test years, while

2021 displayed the most diverse county-level outcomes. A key observation is that the LSTM model's predicted yields consistently exhibited less spatial variability (i.e., smaller standard deviations) than the actual yields across all four test years. This suggests that while the model captures regional mean trends reasonably well, it tends to predict a narrower, more uniform range of yield outcomes across counties than was actually observed. This comprehensive visualization helps in assessing not only the model's accuracy in tracking mean regional yields over time but also its representation of inter-county yield variability.

Understanding these spatial and temporal uncertainty patterns is critical for translating this framework into applied agricultural settings. Deep learning models inherently face challenges when predicting outcomes under unprecedented extreme weather events or agricultural shocks, and purely deterministic models may provide overconfident, inaccurate yield estimates in such cases. Our probabilistic framework offers an important safeguard by explicitly quantifying prediction uncertainty and revealing where the model is least reliable. For a farmer or agronomist, unusually wide prediction intervals or systematically higher uncertainty in certain regions can serve as an automated warning flag, indicating that local expert assessment and on-the-ground scouting are especially important. Rather than serving as a replacement for agronomic expertise, uncertainty-aware machine learning is best used to prioritize where human attention and additional data collection are most critically needed, and to transparently communicate what the model does and does not know. This honesty about model limitations is essential for building trust among growers and decision-makers and for using these tools responsibly under increasing climate-related variability and extremes.

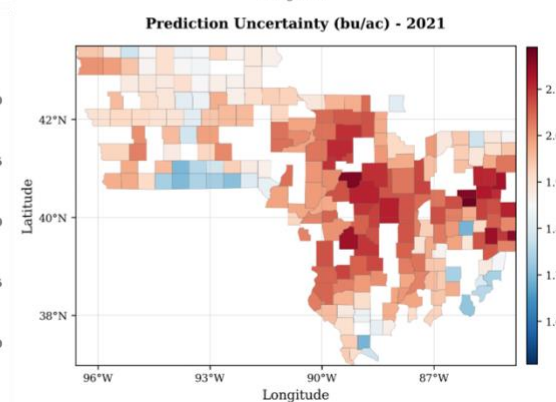
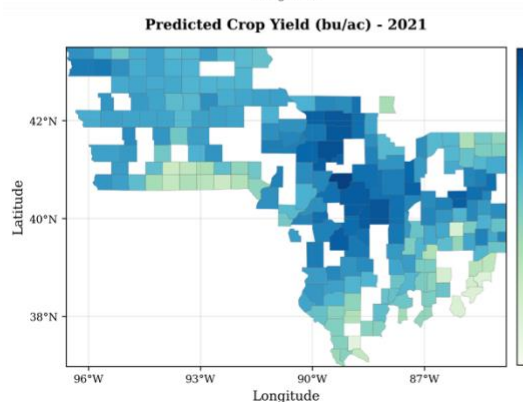
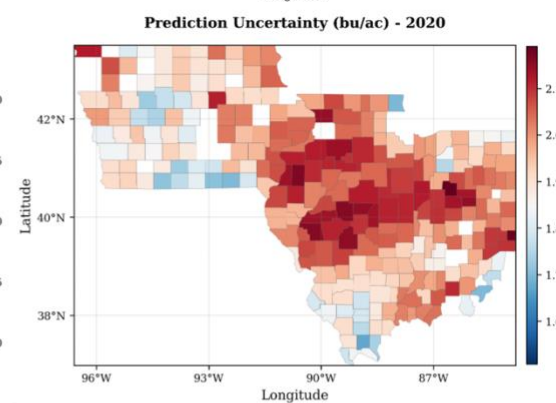
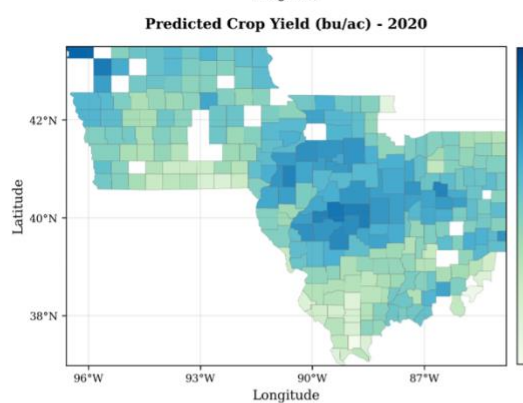
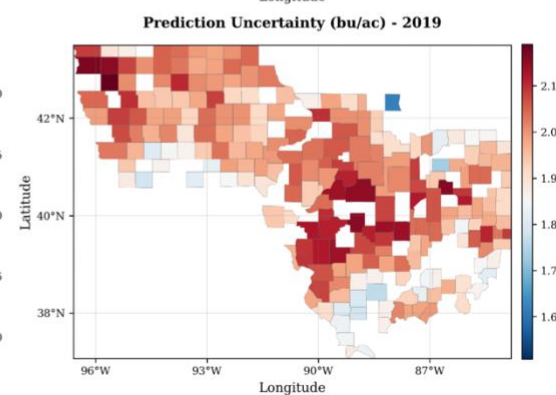
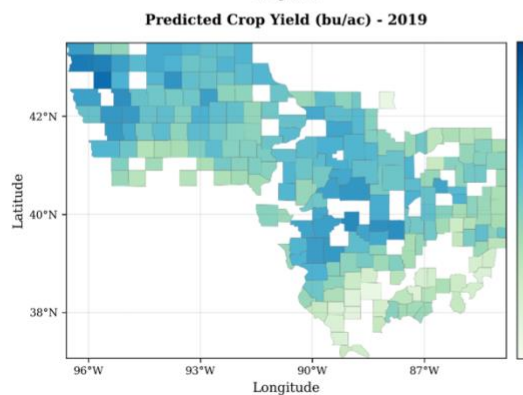
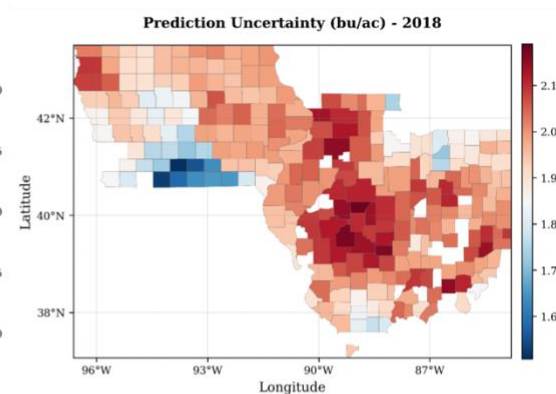
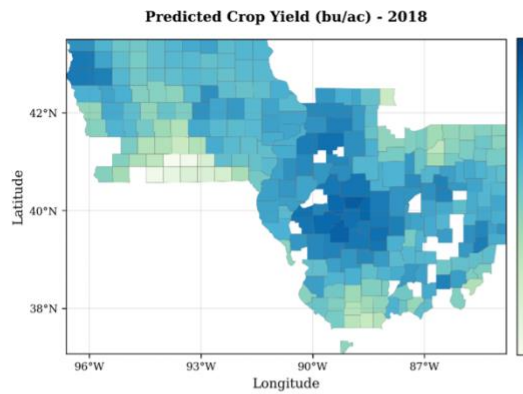


Fig.9. Spatial distribution of LSTM-predicted soybean yields (bu/ac) and associated uncertainties across Illinois, Iowa and Indiana from 2018 to 2021. White areas represent counties not included in the final study analysis or for which data were unavailable.

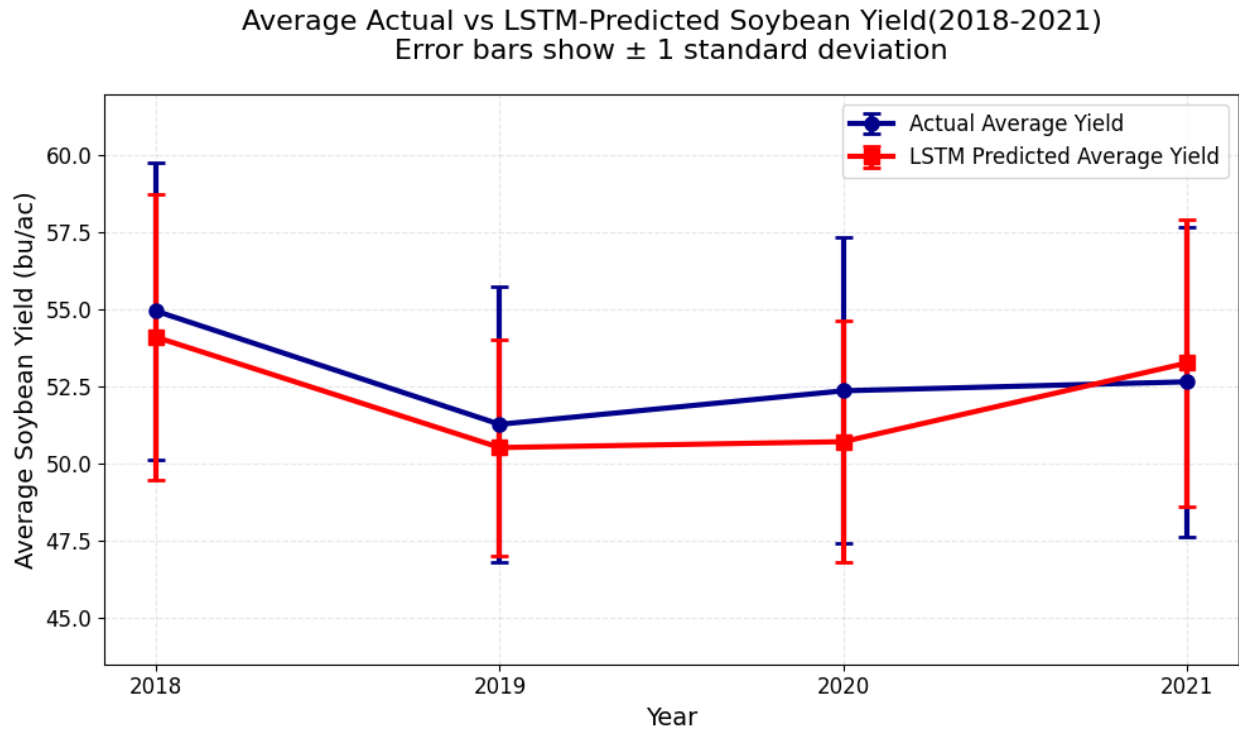


Fig.10. Time series comparison of average actual versus average LSTM-predicted soybean yield (bu/ac) for the test period 2018-2021. Error bars represent ± 1 standard deviation of the county-level yields (for actual) or county-level predicted yields (for LSTM) around the plotted regional mean for each year, indicating the spatial variability across counties.

4.3. Feature and Growth Stage Importance

The permutation importance analysis reveals the relative contribution of different environmental and vegetation variables to soybean yield prediction. As shown in Fig. 11, EVI2 emerges as the most important predictor with a substantially higher importance score compared to other variables. This dominant importance of EVI2 suggests that satellite-derived vegetation condition is the most crucial indicator for yield prediction, likely due to its direct representation of crop growth status and vigor throughout the growing season. Specifically, a higher EVI2 reflects greater photosynthetic activity and canopy health, which are vital for supplying

carbohydrates during critical soybean reproductive phases such as pod development and seed fill (Chlingaryan et al., 2018; Guan et al., 2017). Following EVI2, soil moisture at 10cm depth ranks as the second most important variable, though with a considerably lower importance score. The high importance of shallow soil moisture highlights the significance of root-zone water availability for soybean growth and yield formation. Adequate shallow soil moisture is particularly crucial during soybean flowering to support pod set, and throughout the subsequent seed-filling period to ensure optimal seed development and weight. Deficiencies in this upper soil layer quickly lead to water stress, which can trigger physiological responses like flower and pod abortion or limit assimilate partitioning to the seeds, thereby reducing yield (Jumrani & Bhatia, 2018, 2019). Land surface temperature (LST) during daytime and evapotranspiration (Evap_tavg) also show moderate importance, with scores around 1.5, indicating their relevant but secondary role in yield prediction. Other environmental variables, including deeper soil moisture layers (40cm and 100cm), precipitation (ppt), nighttime land surface temperature (LST_Night), and vapor pressure deficit metrics (vpdmax, vpdmin), demonstrate relatively lower importance scores. This pattern suggests that while these variables contribute to yield prediction, their individual influence is less direct compared to vegetation condition and shallow soil moisture. These findings emphasize the particular value of satellite-derived vegetation indices in crop yield prediction models, while also highlighting the complementary role of soil moisture and thermal conditions.

The analysis of phenological stage importance reveals a clear temporal pattern in their contribution to yield prediction. Fig. 11 shows that the importance of phenological stages generally increases as the growing season progresses, with the late-season stages having the strongest influence on yield prediction accuracy. Stage 6, corresponding to the final reproductive

phase, demonstrates the highest importance score (approximately 0.29), followed by Stage 5 (approximately 0.21) and Stage 4 (approximately 0.13). In contrast, the early stages (Stages 1-3) show minimal importance scores near zero. This pattern suggests that environmental conditions and crop status during later growth stages are more directly related to final yield outcomes compared to early-season conditions. This finding aligns with the biological understanding of soybean yield formation, where the final reproductive stages, including pod-filling and maturation, are critical periods determining grain weight and final yield. The increasing importance of successive growth stages also suggests that the model effectively captures the cumulative nature of crop yield development, where later stages integrate the effects of both current and previous growing conditions.

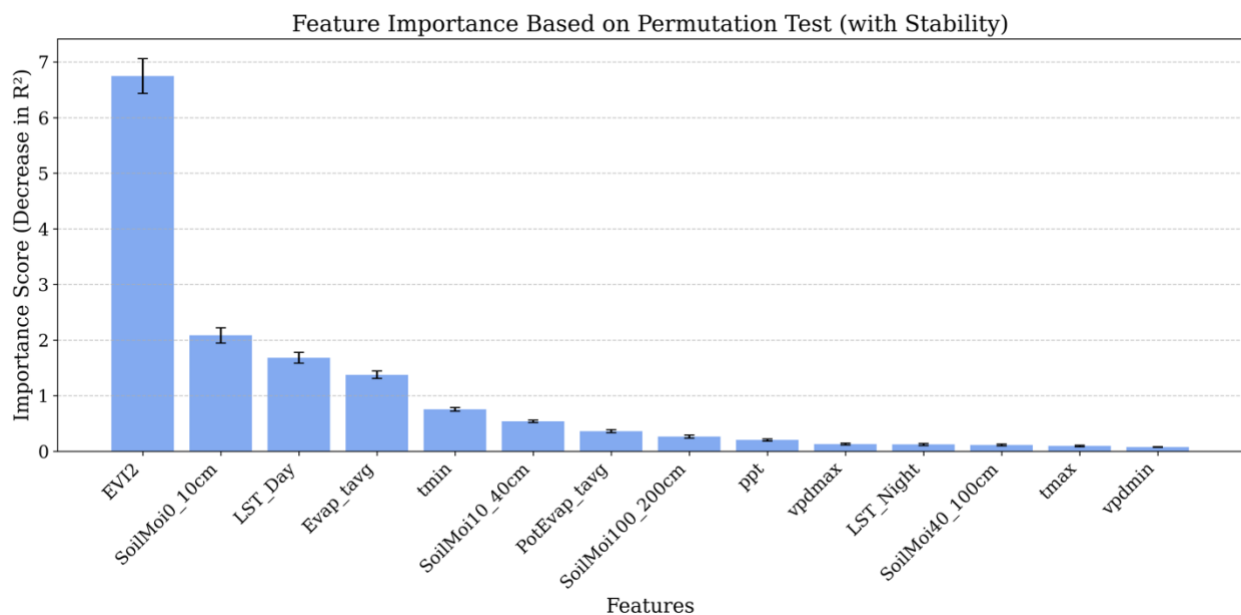


Fig.11. Relative importance of environmental and vegetation variables for soybean yield prediction based on permutation importance test. Bars indicate the mean importance score (decrease in R^2 resulting from feature permutation), while error bars represent ± 1 standard deviation of these scores across 10 permutation repeats, indicating the stability of the importance rankings.

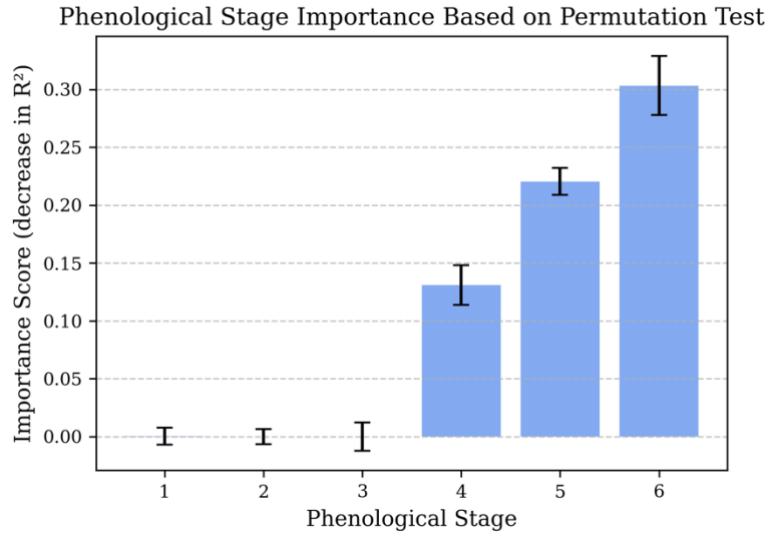


Fig.12. Relative importance of phenological stages for soybean yield prediction based on permutation importance test. Stages progress from early vegetative (Stage 1) to final reproductive phase (Stage 6). Bars indicate the mean importance score (decrease in R^2 resulting from feature permutation), while error bars represent ± 1 standard deviation of these scores across 10 permutation repeats, indicating the stability of the importance rankings.

4.4. Limitations and Future Directions

While this chapter provides a practical demonstration of a phenology-guided, probabilistic framework for soybean yield prediction, several limitations, inherent to the scope of a methodological guide, also highlight avenues for future research. The current county-level analysis does not explicitly incorporate fine-grained agronomic inputs such as specific management practices, maturity group distributions, or precise planting dates, primarily due to data acquisition challenges at this scale. The absence of these critical agronomic drivers may lead to biased predictions in regions with high management heterogeneity, potentially limiting model transferability across different agricultural systems. Although our use of adaptive satellite-derived phenology captures some local crop development variations, integrating more detailed agronomic data could enhance model specificity, especially for finer-resolution studies. Furthermore, the model evaluation focused on temporal hold-out validation within the study region, with results based on reproducible single training runs. A more exhaustive validation of

model robustness would involve broader spatial generalization testing (e.g., on external regions or via leave-one-state-out methods). Before deploying these methods in new regions, we recommend conducting local validation studies and potentially retraining models with region-specific data. These directions represent natural extensions of the current methodology rather than fundamental limitations, positioning this phenology-guided framework as a foundation for continued advancement in agricultural remote sensing applications.

5. Conclusion

This study demonstrates an applied phenology-guided deep learning framework that addresses two critical limitations in current crop yield prediction approaches: the lack of explicit representation of crop phenological stages and the absence of uncertainty quantification. By integrating satellite-derived phenological information with environmental and vegetation time series data, our approach aligns the modeling process with the biological reality of crop development, marking a significant advancement over traditional methods that rely on fixed calendar intervals.

Through comprehensive analysis of soybean yields in Illinois, Iowa, and Indiana, we demonstrated that this phenology-guided approach, when combined with LSTM architecture, achieves robust performance (overall RMSE of 3.49 bu/ac and R^2 of 0.51). The temporal analysis revealed the critical influence of late-season conditions on final yield outcomes, validating the importance of incorporating phenological information. Feature importance analysis identified satellite-derived vegetation index (EVI2) and shallow soil moisture (10cm) as the most influential predictors, highlighting the complementary value of multiple data sources within our phenology-guided framework.

The integration of probabilistic modeling into our framework enabled uncertainty quantification, providing valuable information about prediction reliability across different conditions and locations. Spatial analysis of uncertainties revealed systematic patterns related to yield levels and geographical factors, offering insights into model limitations and providing crucial information for risk assessment. Higher uncertainties were consistently observed in regions along geographical boundaries and in areas with extreme yield predictions, demonstrating the model's ability to identify challenging prediction scenarios.

Our framework integrates domain knowledge of crop phenology with advanced deep learning techniques, providing a data-driven approach that incorporates biological understanding of crop development. This synthesis provides a more comprehensive and biologically informed solution for crop yield prediction. The explicit handling of uncertainties enhances the practical utility and interpretability of predictions, making the approach particularly valuable for agricultural decision-making and policy planning. However, tailoring these outputs for specific end-user applications, such as detailed insurance products or farm-level extension advice, would require further targeted development beyond the methodological demonstration in this chapter.

Looking forward, while the current study demonstrates the framework within a specific region, investigating its adaptability and performance for other crops and in different agricultural regions worldwide presents a promising avenue for future research. As we face increasing challenges in global food security driven by agricultural shocks such as drought and flooding, such advanced tools that combine biological understanding with machine learning capabilities become increasingly critical for supporting sustainable agricultural management.

Acknowledgements

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Reference

- Barbosa, A., Trevisan, R., Hovakimyan, N., & Martin, N. F. (2020). Modeling yield response to crop management using convolutional neural networks. *Computers and Electronics in Agriculture*, 170, 105197. <https://doi.org/10.1016/j.compag.2019.105197>
- Ben Hamida, A., Benoit, A., Lambert, P., & Ben Amar, C. (2018). 3-D deep learning approach for remote sensing image classification. *IEEE Transactions on Geoscience and Remote Sensing*, 56(8), 4420–4434. <https://doi.org/10.1109/TGRS.2018.2818945>
- Bi, L., Wally, O., Hu, G., Tenuta, A. U., Kandel, Y. R., & Mueller, D. S. (2023). A transformer-based approach for early prediction of soybean yield using time-series images. *Frontiers in Plant Science*, 14. <https://doi.org/10.3389/fpls.2023.1173036>
- Bolton, D. K., & Friedl, M. A. (2013). Forecasting crop yield using remotely sensed vegetation indices and crop phenology metrics. *Agricultural and Forest Meteorology*, 173, 74–84. <https://doi.org/10.1016/j.agrformet.2013.01.007>

Butts-Wilmsmeyer, C. J., Seebauer, J. R., Singleton, L., & Below, F. E. (2019). Weather during key growth stages explains grain quality and yield of maize. *Agronomy*, 9(1).

<https://doi.org/10.3390/agronomy9010016>

Carletto, C., Jolliffe, D., & Banerjee, R. (2015). From Tragedy to Renaissance: Improving Agricultural Data for Better Policies. *Https://Doi.Org/10.1080/00220388.2014.968140*, 51(2), 133–148. <https://doi.org/10.1080/00220388.2014.968140>

Chlingaryan, A., Sukkarieh, S., & Whelan, B. (2018). Machine learning approaches for crop yield prediction and nitrogen status estimation in precision agriculture: A review.

Computers and Electronics in Agriculture, 151(June), 61–69.

<https://doi.org/10.1016/j.compag.2018.05.012>

Cohen, I., Zandalinas, S. I., Fritschi, F. B., Sengupta, S., Fichman, Y., Azad, R. K., & Mittler, R. (2021). The impact of water deficit and heat stress combination on the molecular response, physiology, and seed production of soybean. *Physiologia Plantarum*, 172(1), 41–52. <https://doi.org/10.1111/PPL.13269>

Diao, C. (2019). Innovative pheno-network model in estimating crop phenological stages with satellite time series. *ISPRS Journal of Photogrammetry and Remote Sensing*, 153, 96–109. <https://doi.org/10.1016/J.ISPRSJPRS.2019.04.012>

Drusch, M., Del Bello, U., Carlier, S., Colin, O., Fernandez, V., Gascon, F., Hoersch, B., Isola, C., Laberinti, P., Martimort, P., Meygret, A., Spoto, F., Sy, O., Marchese, F., & Bargellini, P. (2012). Sentinel-2: ESA's Optical High-Resolution Mission for GMES Operational Services. *Remote Sensing of Environment*, 120, 25–36.

<https://doi.org/10.1016/j.rse.2011.11.026>

- Ganguly, S., Friedl, M. A., Tan, B., Zhang, X., & Verma, M. (2010). Land surface phenology from MODIS: Characterization of the Collection 5 global land cover dynamics product. *Remote Sensing of Environment*, 114(8), 1805–1816.
<https://doi.org/10.1016/j.rse.2010.04.005>
- Guan, K., Wu, J., Kimball, J. S., Anderson, M. C., Frohling, S., Li, B., Hain, C. R., & Lobell, D. B. (2017). The shared and unique values of optical, fluorescence, thermal and microwave satellite data for estimating large-scale crop yields. *Remote Sensing of Environment*, 199, 333–349. <https://doi.org/10.1016/j.rse.2017.06.043>
- Guo, Y., Fu, Y., Hao, F., Zhang, X., Wu, W., Jin, X., Robin Bryant, C., & Senthilnath, J. (2021). Integrated phenology and climate in rice yields prediction using machine learning methods. *Ecological Indicators*, 120, 106935.
<https://doi.org/10.1016/j.ecolind.2020.106935>
- Han, J., Zhang, Z., Cao, J., Luo, Y., Zhang, L., Li, Z., & Zhang, J. (2020). Prediction of Winter Wheat Yield Based on Multi-Source Data and Machine Learning in China. *Remote Sensing*, 12(2), Article 2. <https://doi.org/10.3390/rs12020236>
- Hüllermeier, E., & Waegeman, W. (2021). Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods. *Machine Learning*, 110(3), 457–506.
<https://doi.org/10.1007/s10994-021-05946-3>
- Jiang, H., Hu, H., Zhong, R., Xu, J., Xu, J., Huang, J., Wang, S., Ying, Y., & Lin, T. (2020). A deep learning approach to conflating heterogeneous geospatial data for corn yield estimation: A case study of the US Corn Belt at the county level. *Global Change Biology*, 26(3), 1754–1766. <https://doi.org/10.1111/gcb.14885>

- Jumrani, K., & Bhatia, V. S. (2018). Impact of combined stress of high temperature and water deficit on growth and seed yield of soybean. *Physiology and Molecular Biology of Plants*, 24(1), 37–50. <https://doi.org/10.1007/S12298-017-0480-5>
- Jumrani, K., & Bhatia, V. S. (2019). Interactive effect of temperature and water stress on physiological and biochemical processes in soybean. *Physiology and Molecular Biology of Plants*, 25(3), 667–681. <https://doi.org/10.1007/S12298-019-00657-5>
- Kamilaris, A., & Prenafeta-Boldú, F. X. (2018). Deep learning in agriculture: A survey. *Computers and Electronics in Agriculture*, 147, 70–90.
- Kang, Y., Ozdogan, M., Zhu, X., Ye, Z., Hain, C., & Anderson, M. (2020). Comparative assessment of environmental variables and machine learning algorithms for maize yield prediction in the US Midwest. *Environmental Research Letters*, 15(6), 064005. <https://doi.org/10.1088/1748-9326/ab7df9>
- Kattenborn, T., Leitloff, J., Schiefer, F., & Hinz, S. (2021). Review on Convolutional Neural Networks (CNN) in vegetation remote sensing. *ISPRS Journal of Photogrammetry and Remote Sensing*, 173, 24–49. <https://doi.org/10.1016/J.ISPRSJPRS.2020.12.010>
- Kendall, A., & Gal, Y. (2017). What uncertainties do we need in Bayesian deep learning for computer vision? *Advances in Neural Information Processing Systems*, 2017-Decem, 5575–5585.
- Khaki, S., Wang, L., & Archontoulis, S. V. (2020). A CNN-RNN Framework for Crop Yield Prediction. *Frontiers in Plant Science*, 10, 1750-. <https://doi.org/10.3389/fpls.2019.01750>
- Koirala, A., Walsh, K. B., Wang, Z., & McCarthy, C. (2019). Deep learning – Method overview and review of use for fruit detection and yield estimation. *Computers and Electronics in Agriculture*, 162, 219–234. <https://doi.org/10.1016/j.compag.2019.04.017>

- Lin, F., Crawford, S., Guillot, K., Zhang, Y., Chen, Y., Yuan, X., Chen, L., Williams, S., Minvielle, R., Xiao, X., Gholson, D., Ashwell, N., Setiyono, T., Tubana, B., Peng, L., Bayoumi, M., & Tzeng, N.-F. (2023). *MMST-ViT: Climate Change-aware Crop Yield Prediction via Multi-Modal Spatial-Temporal Vision Transformer*. 5774–5784. https://openaccess.thecvf.com/content/ICCV2023/html/Lin_MMST-ViT_Climate_Change-aware_Crop_Yield_Prediction_via_Multi-Modal_Spatial-Temporal_Vision_ICCV_2023_paper.html
- Lin, T., Zhong, R., Wang, Y., Xu, J., Jiang, H., Xu, J., Ying, Y., Rodriguez, L., Ting, K. C., & Li, H. (2020). DeepCropNet: A deep spatial-temporal learning framework for county-level corn yield estimation. *Environmental Research Letters*, 15(3). <https://doi.org/10.1088/1748-9326/ab66cb>
- Lobell, D. B., Cassman, K. G., & Field, C. B. (2009). Crop Yield Gaps: Their Importance, Magnitudes, and Causes. *Http://Dx.Doi.Org/10.1146/Annurev.Environ.041008.093740*, 34, 179–204. <https://doi.org/10.1146/ANNUREV.ENVIRON.041008.093740>
- Ma, Y., Zhang, Z., Kang, Y., & Özdoğan, M. (2021). Corn yield prediction and uncertainty analysis based on remotely sensed variables using a Bayesian neural network approach. *Remote Sensing of Environment*, 259, 112408. <https://doi.org/10.1016/j.rse.2021.112408>
- Maimaitijiang, M., Sagan, V., Sidike, P., Hartling, S., Esposito, F., & Fritschi, F. B. (2020). Soybean yield prediction from UAV using multimodal data fusion and deep learning. *Remote Sensing of Environment*, 237, 111599. <https://doi.org/10.1016/j.rse.2019.111599>
- Martínez-Ferrer, L., Piles, M., & Camps-Valls, G. (2021). Crop Yield Estimation and Interpretability With Gaussian Processes. *IEEE Geoscience and Remote Sensing Letters*, 18(12), 2043–2047. <https://doi.org/10.1109/LGRS.2020.3016140>

- Muruganantham, P., Wibowo, S., Grandhi, S., Samrat, N. H., & Islam, N. (2022). A Systematic Literature Review on Crop Yield Prediction with Deep Learning and Remote Sensing. *Remote Sensing*, 14(9), 1990. <https://doi.org/10.3390/rs14091990>
- Nafziger, E. (2019). Corn and Soybean Crops at Mid-Season, 2019. *Farmdoc Daily*, 9(137).
- Rasmussen, M. S. (1992). Assessment of millet yields and production in northern Burkina Faso using integrated NDVI from the AVHRR. *International Journal of Remote Sensing*, 13(18), 3431–3442. <https://doi.org/10.1080/01431169208904132>
- Rippey, B. (2019). Nation’s Wettest 12-Month Period on Record Slows Down 2019 Planting Season. *US Department of Agriculture Blog*.
- Schaaf, C., & Wang, Z. (2015). MCD43A4 MODIS/Terra+ Aqua BRDF/Albedo Nadir BRDF Adjusted Ref Daily L3 Global-500m V006. NASA EOSDIS Land Processes DAAC. *USGS Earth Resources Observation and Science (EROS) Center, Sioux Falls, South Dakota (https://lpdaac.usgs.gov)*.
- Schnitkey, G., & Zulauf, C. (2019). Late planting decisions in 2019. *Farmdoc Daily*, 9(83).
- Schwalbert, R., Amado, T., Nieto, L., Corassa, G., Rice, C., Peralta, N., Schauburger, B., Gornott, C., & Ciampitti, I. (2020). Mid-season county-level corn yield forecast for US Corn Belt integrating satellite imagery and weather variables. *Crop Science*, 60(2), 739–750. <https://doi.org/10.1002/csc2.20053>
- Sharma, R. C., Hara, K., & Hirayama, H. (2018). Improvement of Countrywide Vegetation Mapping over Japan and Comparison to Existing Maps. *Advances in Remote Sensing*, 07(03), 163–170. <https://doi.org/10.4236/ars.2018.73011>

- Sherrick, B. J., Lanoue, C. A., Woodard, J., Schnitkey, G. D., & Paulson, N. D. (2014). Crop yield distributions: Fit, efficiency, and performance. *Agricultural Finance Review*, 74(3), 348–363. <https://doi.org/10.1108/AFR-05-2013-0021/FULL/XML>
- Son, N. T., Chen, C. F., Chen, C. R., Minh, V. Q., & Trung, N. H. (2014). A comparative analysis of multitemporal MODIS EVI and NDVI data for large-scale rice yield estimation. *Agricultural and Forest Meteorology*, 197, 52–64. <https://doi.org/10.1016/J.AGRFORMET.2014.06.007>
- Sun, J., Di, L., Sun, Z., Shen, Y., & Lai, Z. (2019). County-level soybean yield prediction using deep CNN-LSTM model. *Sensors (Switzerland)*, 19(20), 1–21. <https://doi.org/10.3390/s19204363>
- Thornton, M. M., Shrestha, R., Wei, Y., Thornton, P. E., & Kao, S.-C. (2020). Daymet: Daily Surface Weather Data on a 1-km Grid for North America, Version 4. *ORNL DAAC*. <https://doi.org/10.3334/ORNLDAAAC/1840>
- Tian, H., Wang, P., Tansey, K., Zhang, J., Zhang, S., & Li, H. (2021). An LSTM neural network for improving wheat yield estimates by integrating remote sensing data and meteorological data in the Guanzhong Plain, PR China. *Agricultural and Forest Meteorology*, 310, 108629. <https://doi.org/10.1016/J.AGRFORMET.2021.108629>
- USDA. (2022). Usda-national agricultural statistics service, cropland data layer. *USDA-NASS, Washington, DC*.
- van Klompenburg, T., Kassahun, A., & Catal, C. (2020). Crop yield prediction using machine learning: A systematic literature review. *Computers and Electronics in Agriculture*, 177(August), 105709. <https://doi.org/10.1016/j.compag.2020.105709>

- Wang, X., Huang, J., Feng, Q., & Yin, D. (2020). Winter wheat yield prediction at county level and uncertainty analysis in main wheat-producing regions of China with deep learning approaches. *Remote Sensing*, 12(11), 1744. <https://doi.org/10.3390/rs12111744>
- Wang, Z., Schaaf, C. B., Sun, Q., Shuai, Y., & Román, M. O. (2018). Capturing rapid land surface dynamics with Collection V006 MODIS BRDF/NBAR/Albedo (MCD43) products. *Remote Sensing of Environment*, 207, 50–64.
- Weiss, M., Jacob, F., & Duveiller, G. (2020). Remote sensing for agricultural applications: A meta-review. *Remote Sensing of Environment*, 236(August 2019), 111402. <https://doi.org/10.1016/j.rse.2019.111402>
- Westcott, P. (2010). *USDA agricultural projections to 2019*. DIANE Publishing.
- Williams, C. K., & Rasmussen, C. E. (2006). *Gaussian processes for machine learning* (Vol. 2). MIT press Cambridge, MA.
- Xie, W., Zhao, M., Liu, Y., Yang, D., Huang, K., Fan, C., & Wang, Z. (2024). Recent advances in Transformer technology for agriculture: A comprehensive survey. *Engineering Applications of Artificial Intelligence*, 138, 109412. <https://doi.org/10.1016/j.engappai.2024.109412>
- You, J., Li, X., Low, M., Lobell, D., & Ermon, S. (2017). Deep Gaussian process for crop yield prediction based on remote sensing data. *31st AAAI Conference on Artificial Intelligence, AAAI 2017*, 4559–4565. <https://doi.org/10.1609/aaai.v31i1.11172>
- Zhang, C., & Diao, C. (2023). A Phenology-guided Bayesian-CNN (PB-CNN) framework for soybean yield estimation and uncertainty analysis. *ISPRS Journal of Photogrammetry and Remote Sensing*, 205, 50–73. <https://doi.org/10.1016/J.ISPRSJPRS.2023.09.025>

- Zhong, L., Hu, L., Zhou, H., & Tao, X. (2019). Deep learning based winter wheat mapping using statistical data as ground references in Kansas and northern Texas, US. *Remote Sensing of Environment*, 233(August), 111411. <https://doi.org/10.1016/j.rse.2019.111411>
- Zhu, X. X., Tuia, D., Mou, L., Xia, G.-S., Zhang, L., Xu, F., & Fraundorfer, F. (2017). Deep learning in remote sensing: A comprehensive review and list of resources. *IEEE Geoscience and Remote Sensing Magazine*, 5(4), 8–36.